

7. Find the maximal area of a right triangle with hypotenuse of length 1.

$$A = \frac{1}{2}xy \quad x^2 + y^2 = 1$$

$$A = \frac{1}{2}x(\sqrt{1-x^2})$$

$$A' = \frac{1}{2}x\left(\frac{1}{2}(1-x^2)^{-\frac{1}{2}} \cdot (-2x)\right) + \frac{1}{2}\sqrt{1-x^2}$$

$$1-x^2=0$$

$$-\frac{1}{4}2x^2 \cdot (1-x^2)^{-\frac{1}{2}} + \frac{1}{2}\sqrt{1-x^2}$$

$$A = \frac{1}{2}x(\sqrt{1-x^2})$$

$$A = \frac{1}{2}\sqrt{x^2 - x^4}$$

$$A' = \frac{1}{2} \cdot \frac{1}{2}(x^2 - x^4)^{-\frac{1}{2}} \cdot (2x - 4x^3)$$

$$\frac{1}{4} \cdot \frac{2x - 4x^3}{\sqrt{x^2 - x^4}} = 0$$

$$2x - 4x^3 = 0$$

$$x(2 - 4x^2) = 0$$

$$2 - 4x^2 = 0 \quad x = \pm \sqrt{\frac{1}{2}}$$

$$\sqrt{\frac{1}{2}}$$

$$A = \frac{1}{2} \cdot \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2}} = \frac{1}{4}$$

$$23. \int x \ln(x-1) dx$$

$$u = \ln(x-1)$$

$$dv = x dx$$

$$\frac{x^2}{2} \ln(x-1) - \int \frac{x^2}{2} \cdot \frac{1}{x-1} dx \quad v = \frac{x^2}{2}$$

$$\frac{x^2}{2} \ln(x-1) - \int \frac{1}{2} \frac{x^2}{x-1} dx$$

15. $\lim_{x \rightarrow 0} \frac{\sin(2x)}{\sin(3x)}$

$$\frac{\sin(0)}{\sin(0)} = \frac{0}{0}$$

$$\frac{2 \cos(2x)}{3 \cos(3x)} = \frac{2}{3}$$

$$14 \rightarrow 11 - 9.5 \rightarrow 8.75 \rightarrow 8.375$$

$$X \cdot 0.5 + 4$$

18	8.625
7	7.968
20	8.75
29	8.3135
37	9.8125
26	8.562
22	8.4375
36	9.75
2	7.625

$$8 \dots 5 \rightarrow 4 + 4 \rightarrow 8$$

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$$1.01x + 75$$

500	108
200	512
530	856.
125	434.6
25	408.85
600	428.89
420	741...
700	1032.95

$$1.5x - 6$$

$$1.5x - 6 = x$$

$$12$$

17	37.3125
8	-8.35
41.5	-44.95
20.5	55.03
40	153....
16	32.5
42.006	239.86
24	72.75
7	-13.31

$$0.6x - 1$$

$$0.6x - 1 = x$$

$$-1 = 0.4x$$

$$\frac{-1}{0.4} = x$$

$$-2.5$$

169	10.841
5	-1.528
9	-1.0096
3	-1.78
20	.416
15	-0.232
4	-1.99
8	-1.68
12	-0.628

$$y = -x + 6$$

$$2 \rightarrow 4 \rightarrow 2 \rightarrow 4 \rightarrow 2$$

$$-0.5x + 6$$

$$2 \rightarrow 5 \rightarrow 3.5 \rightarrow$$

$$0.5$$

$$3(ans)(1-ans)$$

$$0.5, 0.75, 0.5625, 0.738, 0.571$$

$$3.3(x)(1-x)$$

$$0.5 \rightarrow 0.825 \rightarrow 0.176 \rightarrow 0.825$$